

Machine learning-based prediction of surface structure induced by laser surface texturing

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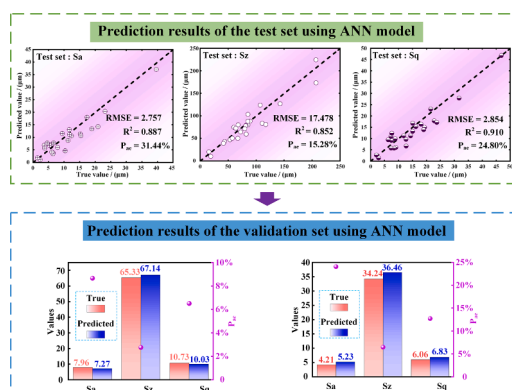
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HIGHLIGHTS

- A surface roughness prediction model exhibiting high accuracy was developed.
- Effect of laser parameters on periodic surface micro/nanostructures was studied.
- Nonlinear correlation between laser parameters and surface roughness was discovered.
- The model shows good generalization capability for diverse parameter sets.

GRAPHICAL ABSTRACT



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ABSTRACT

Surface roughness is a key indicator for the evaluation of laser surface functionalization since it can greatly affect the surface functions including surface wettability, friction and wear properties, optical properties and etc. However, predicting the surface roughness of laser-ablated surfaces for specific laser processing parameters presents a major challenge due to the instantaneity and inherent complexity of laser-material interactions. Traditionally, achieving a desired surface roughness necessitates designing and experimentally testing a vast number of laser parameter combinations, which can be quite time and resource consuming. Machine learning model is capable to extract the potential relations among the raw data based on the data itself and utilizes this for prediction and analysis of surface roughness, which is quite different from the physical models that directly establishes the relationship between the laser processing parameter and surface roughness. This work employs two different models, e.g., LASSO (Least absolute shrinkage and selection operator) and ANN (Artificial Neural Network), for prediction and analysis of the surface roughness on the laser ablated zirconia ceramics surface. The

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experimental results demonstrate that the prediction results for the roughness values (S_a , S_z , S_q) using the ANN model is significantly better than those using the LASSO model, indicating the non-linear relationship between the laser processing parameters and surface roughness. On the validation set, the ANN model achieved RMSE values of 2.757 (S_a), 17.478 (S_z), and 2.854 (S_q), with corresponding R^2 values of 0.887, 0.852, and 0.910. Notably, the model maintained strong performance even for laser processing parameters beyond the pre-set process window, exhibiting prediction errors around only 10 %. In addition, this work confirms that the formation of regular periodic surface micro/nanostructures is critically dependent on specific laser parameter combinations, particularly the requirement for high laser power to be coupled with high scanning speed. The superior generalization ability of the developed model can provide key guidance for the design and optimization of laser processing parameters utilized for the laser surface functionalization of different materials.

1. Introduction

Laser surface texturing technique can induce fine periodic micro/nanostructures on surface via ablation effect and has been widely adopted for design and fabrication of various functional ceramic materials by various researchers [1]. This type of micro/nanostructures is considered the key factor for achieving diverse surface functionalities, such as superhydrophobicity, anti-corrosion and anti-bacterial capability [2]. Zirconia ceramics has been widely applied in the areas including dental implant [3], orthopedic implant [4], coating [5] and optical device [6] due to its superior mechanical strength, biocompatibility and chemical stability. The performance in these applications often demands distinct surface functionalities, which are heavily dependent on precise control of surface roughness. For instance, the surface wettability of zirconia ceramics is the key for modulating the biocompatibility and coating adhesion strength, and the surface wettability is strongly correlated with the surface roughness. However, due to the instantaneity and diversity (e.g., evaporation and ionization of the material) of the laser ablation process, it makes the coupling relationship between the laser processing parameters and surface roughness very complex, which can be very hard to predict.

For traditional prediction model, the relationship between the laser energy and ablation threshold of material surface has to be established first in order to predict the surface roughness of laser textured surface. In addition, the variation process of the internal melt pool during laser-material ablation has to be taken into consideration as well. However, the variation of melt pool during laser ablation process is both dynamic and instantaneous [7]. Moreover, the morphological characteristics of the melt pool, including the shape, size and depth will also greatly affect the processing effect of laser ablated surface, and thus exert influence on surface roughness [8]. This makes it extremely difficult to predict the surface roughness of the laser ablated surface via pure physical models. Indeed, the most straightforward method is to design and conduct experiments using the combination of different laser processing parameters and further investigate their effects on surface roughness. Undoubtedly, this kind of method will be time-consuming and waste large amount of resources. With the development of artificial intelligence, the data-driven machine learning model can process and analyze a great amount of data, and subsequently extract the potential correlation among the raw data. This renders that the machine learning method has been more and more widely used for the prediction of surface characteristics rather than simply relying on theory [9–12]. For instance, Jiang et al. [13] established six contact angle prediction models based on machine learning and realized the prediction for the contact angle of a liquid metal with a solid oxide. The prediction accuracy was as high as 96 %. Kats et al. [14] proposed a physics-informed neural network model to identify the relationship between the local thermal feature and the grain structure characteristics, and features including the average grain size and average grain aspect ratio could be accurately predicted. Ye et al. [15] combined data-driven deep learning with energy deposition model to predict the morphological types of femtosecond LIPSS. The results indicated that by extracting the data features using the deep learning model, the accuracy of self-correlation validation could reach as high as 98.0 %. Zhang et al. [16] established

the prediction model of the melt pool morphology from the built commands in a laser-based powder bed fusion (L-PBF) process. The prediction model utilized long short-term memory (LSTM) network and Melt Pool Generative Adversarial Network (MP-GAN) to synthesize the images and infer the shape of the melt pool. The results demonstrated that an accuracy of 90.7 % could be achieved in estimating the area of the melt pool. Sun et al. [17] built three machine models including support vector regression (SVR), Gaussian process regression (GPR), and artificial neural network (ANN) in order to achieve wideband fatigue life prediction. With the help of sufficient Monte Carlo numerical simulations, it was demonstrated that the developed machine learning models could accurately predict the fatigue life and outperformed the frequency-domain models in terms of the life prediction accuracy. The above-mentioned research progresses clearly shows that the data-driven machine learning model are capable to realize the prediction of key surface characteristics, including surface wettability, surface morphology, melt pool shape and etc. The above-mentioned research progresses also indicates that extensive research progress has been achieved in this area. While for laser ablation, the laser-material interaction can be very complex, and the key underlying mechanism for how different laser processing parameters can induce the regular surface structure should be clearly elucidated. In the meantime, it is also of vital importance to investigate the non-linear correlation between the laser processing parameters and laser-induced surface roughness, which can help to further enhance the prediction accuracy of the machine learning model.

This work utilized exhaustive experiment method to obtain a group of 125 laser textured samples processed by different scanning speeds, pitches and laser powers (5 values for each parameter). The selected roughness parameters for each sample were determined as S_a , S_q and S_z . Next, machine learning-based models including LASSO (Least absolute shrinkage and selection operator) and ANN (Artificial Neural Network) were applied for prediction of surface roughness induced by laser surface texturing, and the performance of the two different models was analyzed and discussed. The experimental results indicated that all the selected laser processing parameters exhibited notable influence on surface roughness. In addition, realization of regular periodic surface micro/nanostructures are largely based on the careful determination of the combination of laser processing parameters.

2. Methods

2.1. Experimental details

The samples used in the experiments are white bulk yttria-stabilized zirconia ceramics (3Y-TZP) manufactured by isostatic pressing purchased from Taizhou Jingwei Special Ceramics, and the purity of the zirconia ceramics is higher than 94.4 %. The laser surface texturing experiments employed a nanosecond laser (MOPA, YDFLP-E2-60-M7-M-R, Shenzhen JPT Opto-electronics Co., Ltd.) with the wavelength of 1064 nm. The zirconia ceramics was ultrasonically cleaned in ethanol and deionized water successively for 5 min before laser ablation to remove surface impurities, and then placed in the air to dry. After that, the surfaces were laser ablated and periodic surface micro/

nanostructures were subsequently induced by laser ablation. As shown in Fig. 1, the laser beam was emitted from the laser source and passed through the isolator, pre-amplifier and main amplifier, and finally entered laser scan head. The scanning area was set as a square with the dimension of 3×3 mm, and a pre-set cross-hatch scanning pattern was utilized and the area scan was repeated for six times. During laser surface scanning, the frequency was fixed at 60 kHz. Three groups of laser processing parameters were investigated in this work: scanning speed (10 mm/s, 50 mm/s, 100 mm/s, 150 mm/s, 200 mm/s), laser power (3.43 W, 5.14 W, 6.86 W, 8.57 W, 10.29 W) and scanning pitch (100 μ m, 150 μ m, 200 μ m, 250 μ m, 300 μ m), and thus sample surfaces with different micro/nanostructures were obtained. More details of the laser processing parameters are shown in Table 1. For surface characterizations, scanning electron microscopy (SEM, Quanta 3D FEG, FEI) was used to observe the morphology and micro/nanostructures on the sample surfaces. Energy-dispersive X-ray spectroscopy (Nova Nano 450) was utilized to reveal the change of surface chemical compositions. Digital microscope (Keyence VHX-6000) was employed for characterization of surface profile and measurement of surface roughness. The surface roughness measurement standard utilized in this work was ISO 25178. During each measurement, three different points of the surface were selected for measurement, and the average value was taken as the final result. The measurement region was selected at a magnification of $500\times$, and its area is 0.3433 mm^2 .

2.2. Machine learning models

Exhaustive experiments were employed in this work and 125 ($5 \times 5 \times 5$) samples were produced. To simultaneously satisfy the requirements of training and test, the 125 samples were randomly divided into two groups. One group with 100 samples was used as the training set for the machine learning model, and the rest of the 25 samples was taken as the test set or validation set. Before the machine learning model

Table 1

Details of the laser processing parameters used in this work.

Parameters	Value
Laser wavelength (nm)	1064
Repetition rate (kHz)	60
Scanning speed (mm/s)	10, 50, 100, 150, 200
Scanning pitch (μ m)	100, 150, 200, 250, 300
Average power (W)	3.43, 5.14, 6.86, 8.57, 10.29
Single pulse energy (mJ)	57.17, 85.67, 114.33, 142.83, 171.5
Spot diameter (μ m)	~ 50

was trained, StandardScale method was utilized to standardize the data of training set and rendered its average value of 0 and standard deviation of 1. This could help to eliminate the influence brought by dimensional difference and features with wide range of value (e.g., S_z). In addition, the distribution of the data would follow the same criterion after standardization which could accelerate the convergence procedure of the model. To prevent the data leakage of the model during the training and test processes, statistical magnitude of the training set was utilized to standardize the test and conduct the subsequent test and validation procedures [18]. The linear model LASSO [19] and ANN model [20] were utilized for the prediction of the laser-induced surface roughness values (S_a , S_z , S_q) and the model performance was evaluated. The relationship between the laser processing parameters and laser-induced surface roughness was analyzed as well. In the LASSO model, the loss function was consisted of least squares term and regularization term (this will be explained with more details in the next section). Coordinate descent was utilized as the optimization algorithm [21], and only one coefficient was optimized for each iteration which would be updated gradually until the model converged. For the ANN model, the loss function was chosen as MSE (Mean Squared Error), which is one of the most commonly used loss functions for evaluation of the difference between the prediction value of the model and the actual

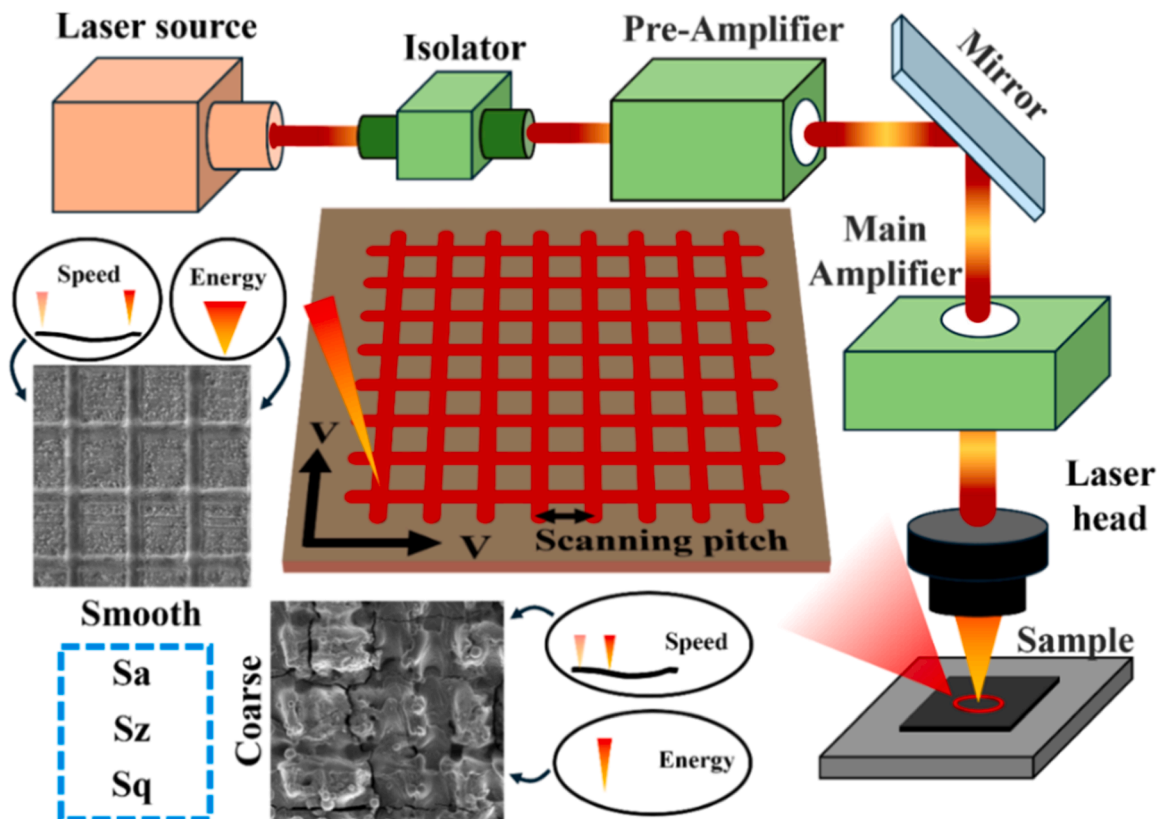


Fig. 1. Process schematic for laser surface texturing of zirconia ceramics.

value. The optimizer was chosen as adaptive moment estimation (ADAM), which combined the merits of the momentum optimizer and root mean square propagation (RMSprop) optimizer and could adaptively adjust the learning rate.

LASSO model is capable to constrain the model parameters by introducing the penalty term of L1 norm into the loss function and thus feature selection and reduction of overfitting can be realized. The mathematical expression of the LASSO model is shown in Eq. 1:

$$Cost(w) = \sum_{j=1}^N (y_i - w^T x_i)^2 + \lambda ||w||_1 \quad (1)$$

LASSO is a powerful linear regression model that can help to establish more concise interpretable model. In the LASSO model, in order to adequately train and test each data point, 10-cross validation method can be introduced and employed [22]. For this method, the data is evenly and randomly distributed into ten subsets. For every iteration, nine subsets are regarded as the training set and the last one is taken as the test set. To validate the results obtained by the 10-cross validation method, new validation set (data set without training and test) can be added to validate the prediction capability of the linear model. By using this method, the generalization ability of the LASSO model can be comprehensively and effectively evaluated. In addition, Gridsearch method is utilized to select and optimize the hyperparameters of the model [23].

ANN model processes the data by imitating the structure and function of the biological neural network, and it is composed of a large number of interconnected nodes (i.e., nerve cell), including input layer, one or many hidden layers and output layer. In this work, dimensionality ascendent and dimensionality reduction of the features can be achieved using the linear layer function, and Relu function is added to introduce the non-linear parameters among the feature in order to realize the prediction capability of the non-linear model. In the meantime, the difference for the number of connection layer is also taken into consideration and its effect on the prediction capability of the model is systematically analyzed. Finally, the optimal number of connection layer is determined for the ultimate model prediction.

2.3. Evaluation indicators

In this work, three types of evaluation indicators are introduced to assess the generalization capability of the machine learning models, including root-mean-square error (RMSE), absolute error (AE) and R^2 . RMSE (Eq. 2) is used to evaluate the fluctuation of the predicted value, i.e., the extent for the deviation of the predicted value from the actual value. It can be very sensitive to the abnormal value. AE (Eq. 3) provides the measurement for the average value, which indicates the average difference between the predicted value and actual value, and can be quite stable. R^2 (Eq. 4) represents the fitting degree of the model to the data, which can demonstrate the ratio of the variability of model interpretation in the total variability, i.e., the interpretability of the

model to the data [24].

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$AE = |y_i - \hat{y}_i| \quad (4)$$

3. Results and discussions

3.1. Surface morphology

Surface morphology with regular periodic micro/nanostructures was formulated as a result of laser ablation, and laser processing parameters could drastically affect the quality of the structural parameters, e.g., regularity, groove width, groove depth and pitch. It can be clearly found that using the same scanning speed, low laser power induced shallow grooves while the surface exhibited relatively high regularity and smoothness. However, high laser power could help to generate deeper grooves and a significant number of peaks and valleys brought by the deposition of ablated particles were formulated on the surface leading to the reduction of surface flatness. Besides laser power, it can be observed from the SEM images that low scanning speed resulted in the formation of more cracks on the zirconia ceramics surface and thus led to the reduction of laser ablated structural quality. This was mainly attributed to the distinct thermal accumulation when the scanning speed was set at a low value [25]. Under this circumstance, the temperature gradient on the zirconia ceramics surface will be highly unbalanced and thus resulted in the uneven stress distribution [26]. When the laser power remained the same, high scanning speed could facilitate the formation of regular periodic micro/nanostructures. The distance between every two grooves was almost identical to the pre-set pitch, and the depth of all the laser ablated grooves turned out to be uniform. In addition, as observed from Fig. 3, low scanning speed will result in irregular surface profile when the laser power is set constant, and the maximum concave depth can reach 60 μm . This is mainly due to the fact that more laser pulses will act on the same region of the zirconia ceramics surface per unit time when the scanning speed is low, and the heat accumulation effect will be more significant [27]. The analysis result of EDS also confirms that the percentage of oxygen is as high as 31.4 % indicating that the zirconia ceramics surface has been excessively ablated. When the scanning speed is set constant (Fig. 3a and c), the irregularity of the zirconia ceramics surface is further increased with higher laser power, and the maximum concave depth and oxygen content are increased to 180 μm and 35.0 % respectively. At this time, the scanning speed requires further increase in order to facilitate the formation of regular micro/nanostructures (as shown in Fig. 3d). It can thus be summarized that both high laser power and high scanning speed are required to formulate regular periodic

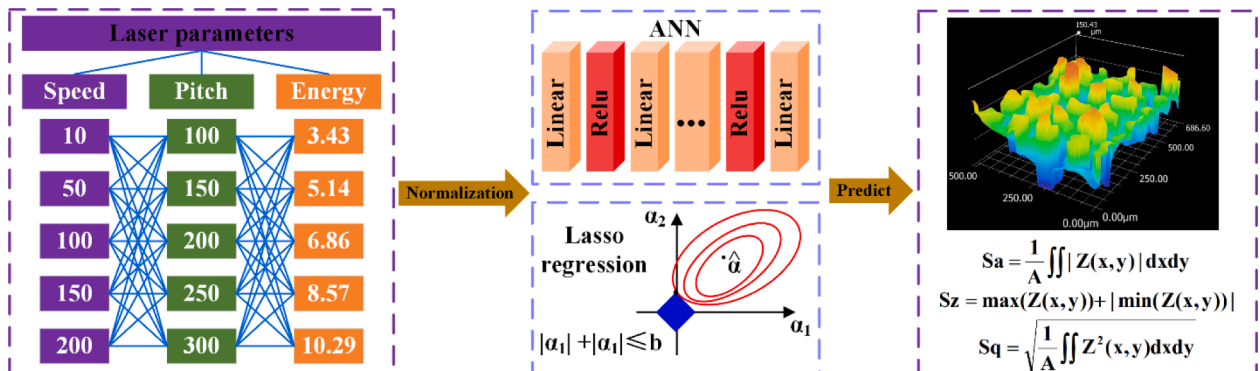


Fig. 2. Flow diagram for prediction of surface roughness.

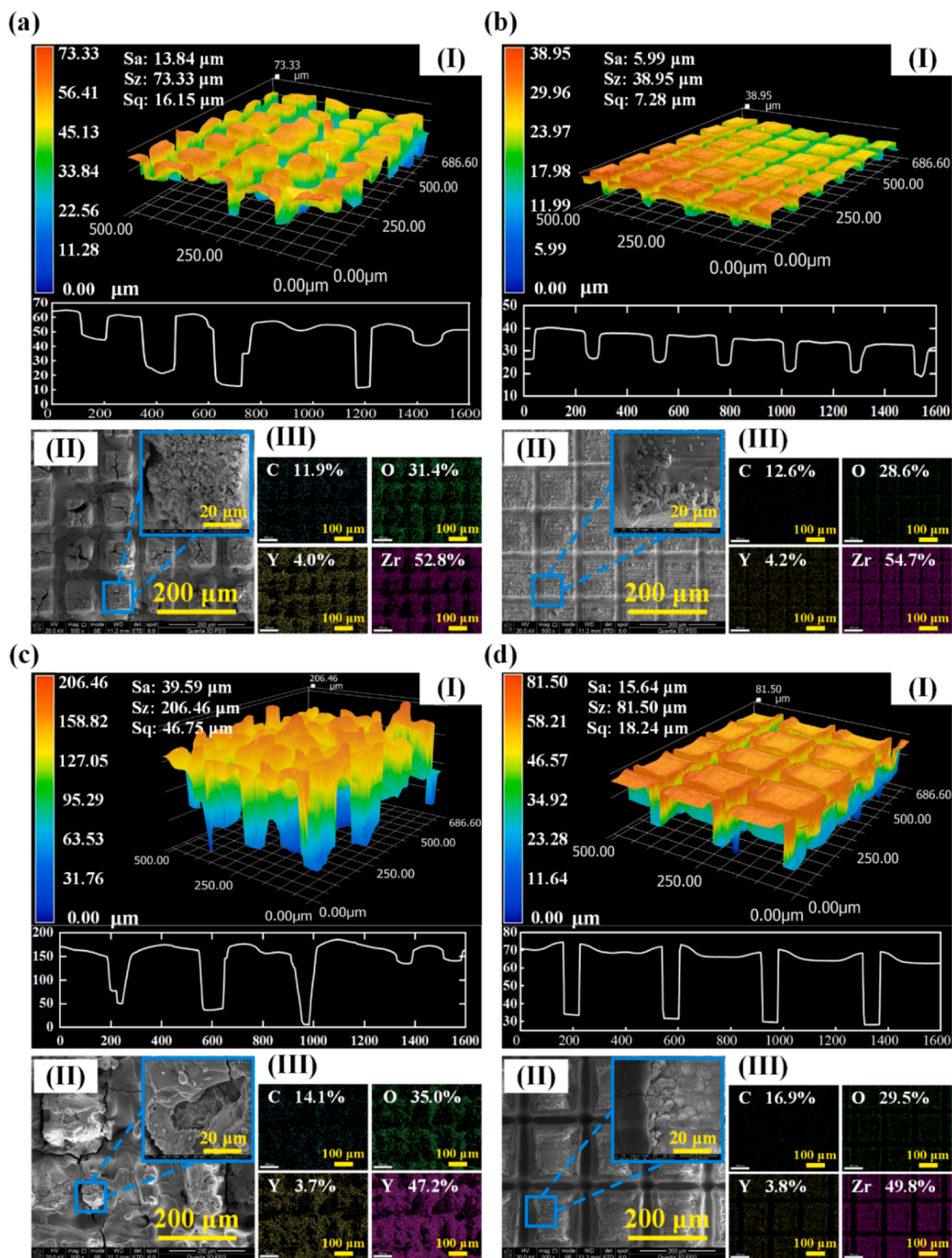


Fig. 3. Surface morphology of the laser textured zirconia ceramics processed by different laser processing parameters: (a) Pitch of 100 μm , scanning speed of 10 mm/s, power of 3.43 W; (b) Pitch of 100 μm , scanning speed of 100 mm/s, power of 3.43 W; (c) Pitch of 150 μm , scanning speed of 10 mm/s, power of 10.29 W; (d) Pitch of 150 μm , scanning speed of 200 mm/s, power of 10.29 W.

surface micro/nanostructures.

3.2. Prediction and validation results of LASSO model

According to the machine learning method and data partitioning method as previously discussed, LASSO model was firstly utilized for

prediction and analysis of the roughness results. Fig. 4 shows the prediction results of three roughness values obtained by the LASSO model. It can be found that using the LASSO model, the prediction results of the three roughness values deviate distinctly from the actual measurement results, and negative prediction results even occurred in some certain ranges. This implies that the LASSO model is incapable to learn and

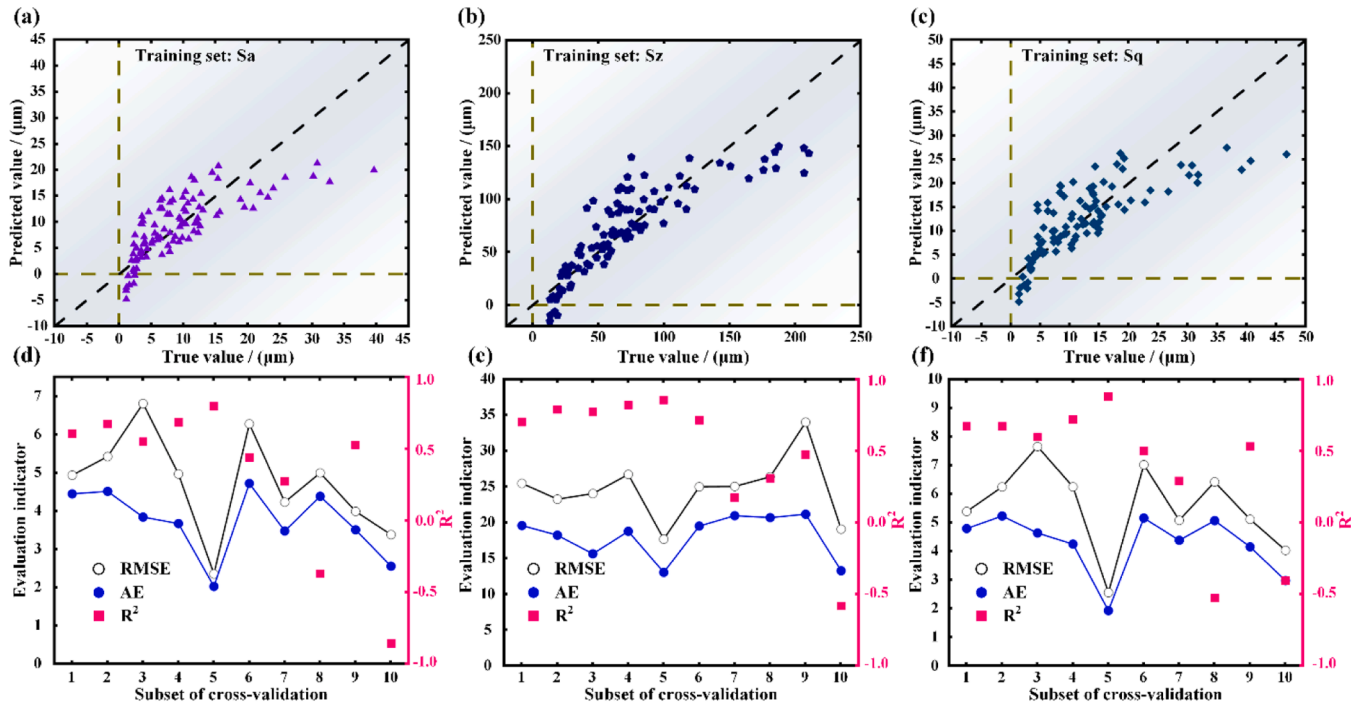


Fig. 4. The actual and predicted surface roughness values obtained by the LASSO model: (a) the results of S_a ; (b) the results of S_z ; (c) the results of S_q ; Analysis for the evaluation indicators of model prediction: (d) the results of S_a ; (e) the results of S_z ; (f) the results of S_q .

comprehend the relationship between the laser processing parameters and surface roughness. While for those data at low roughness values, the prediction results present good agreements with the actual results, showing that the LASSO model exhibits relatively satisfying prediction capability for these data. To further analyze the model performance, the evaluation indicators of model prediction for every subset were discussed as shown in Fig. 4d-f. It can be observed the prediction results for some subsets appears quite accurate, especially for Subset 5, in which the RMSE value (2.36) and AE value (2.03 μm) are relatively low, and the R^2 value is close to 0.9. However, the prediction results for other subsets lacks accuracy and the R^2 values for Subset 8 and Subset 10 are negative. This indicates that the effect of model prediction is not satisfying, which is even worse than simply predicting the average value of the dependent variables. The existence of significant difference among the prediction capability of various subsets is attributed to the error brought by the random assortment data. Nevertheless, this is acceptable since it can reduce the impact caused by data diversity to the minimum [28].

In addition, the LASSO model was utilized for prediction and analysis of the data that has not been trained and tested, as shown in Fig. 5. It can be seen that the prediction accuracy of the LASSO model for S_z is obviously higher than those of S_a and S_q , and the majority of the

predicted values lie on the line $y = x$. The R^2 value for S_z prediction is as high as 0.897, indicating that the model exhibited high fitting degree for the data. Furthermore, the P_{ae} (percentage of absolute error) value reaches 20.29 %, which is indeed relatively small due to the large values of S_z of the data. Although the RMSE indicators of S_a and S_q are only as high as 4, this does not represent that the model shows good fitting degree for these two values since the R^2 value is only around 0.4–0.5. Moreover, their P_{ae} values reach 57.23 % and 41.48 % respectively, indicating that significant difference exists between the predicted values and actual values.

S_z is the height difference between the peak and valley of the laser textured surface, representing the maximum depth of the surface. The LASSO model exhibits distinctly better prediction results for S_z which is likely due to the fact that the linear dependence between the laser processing parameters and S_z is stronger than those of S_a and S_q .

3.3. Prediction and validation results of ANN model

The above-mentioned analysis in terms of the prediction results by the LASSO model indicates that a simple linear model is incapable to accurately predict the relationship between the laser processing parameters and surface roughness. Therefore, this work utilizes artificial

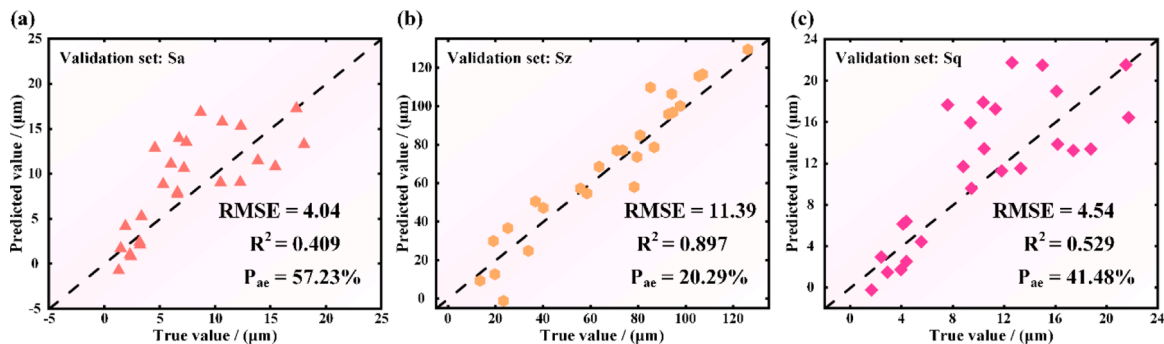


Fig. 5. The analysis of prediction results for the validation sets obtained by the LASSO model: (a) the results of S_a ; (b) the results of S_z ; (c) the results of S_q .

neural network (ANN) and evaluates its prediction capability. ANN is a typical mathematical model that imitates biological neural network, and has been widely utilized in the fields including machine learning and can effectively deal with non-linear problems [29]. By introducing linear layers and activation functions, the ANN model as shown in Fig. 2 is established which can be used for the prediction of surface roughness. Training rounds refer to the number that the whole training set accomplishes the complete process consisting of forward propagation and back propagation. It is an important parameter for the ANN model and has significant impact for the model performance. When the number of training rounds is excessive, the “overfitting” phenomenon will occur due to the excessive fitting of the training data. This will lead to the superior prediction results for the training set using the model but the prediction results for the validation set will be very inaccurate and very sensitive to the abnormal value of data [30]. When the number of training rounds is too low, the “underfitting” phenomenon will occur due to insufficient training. At this time, the fitting capability of the model towards the data of the training set and validation set is relatively poor [31]. Therefore, reasonable control of the number of training rounds is the key to ensure the superior generalization capability of the model. Based on it, this work discusses the relationship between the loss and R^2 value of the model and the number of training rounds, as shown in Fig. 6. With the increase of the training Epoch number (iteration number), the loss of the model exhibits distinct decreasing trend, demonstrating that the prediction capability of the model has been continuously enhanced. It can be clearly found that the loss of the test set is higher than that of the training set. This appears to be normal since the model is fitted on the training set. After an iteration of 800 times, the loss

of the model gradually tends to stabilize, while the R^2 value is relatively low. Therefore, the iteration number is determined as 1000 to ensure the relatively satisfying prediction capability and stability of the model. Moreover, it is observed that the R^2 value of the model is relatively low when the number of training rounds is insufficient. This corresponds to the process that the model search for optimal solution. With the increase of training rounds, the R^2 values gradually become close to 1, indicating the constant enhancement of the fitting degree of the model towards S_a (S_z , S_q). Under this condition, the model can well learn the potential relationship between the input parameters and output parameters. Meanwhile, it can be observed that the R^2 values exhibits certain fluctuations when the number of training rounds are set at 700 or 1100. This should be mainly attributed to the optimization process of the optimizer. During the learning process, the model will constantly iterate and update in order to find out the optimal solution under the action of specific learning rate. If the R^2 value exhibits downtrend, the solution obtained by the model during the iteration process might be local optimal solution rather than global optimal solution [32].

After the number of training rounds was determined, the corresponding ANN model was utilized to predict the surface roughness values of the training set and test set. Fig. 7a-c shows the prediction results and performance indicators for the training set of 100 samples. It can be seen that the predicted values for the majority of the samples are close to the line $y = x$, especially for S_z . This indicates that the predicted values of the samples in the training set match well with the actual values during the training process, and the model can effectively capture the relations among different roughness parameters. To be specific, the R^2 values for all the three roughness parameters are higher than 0.95,

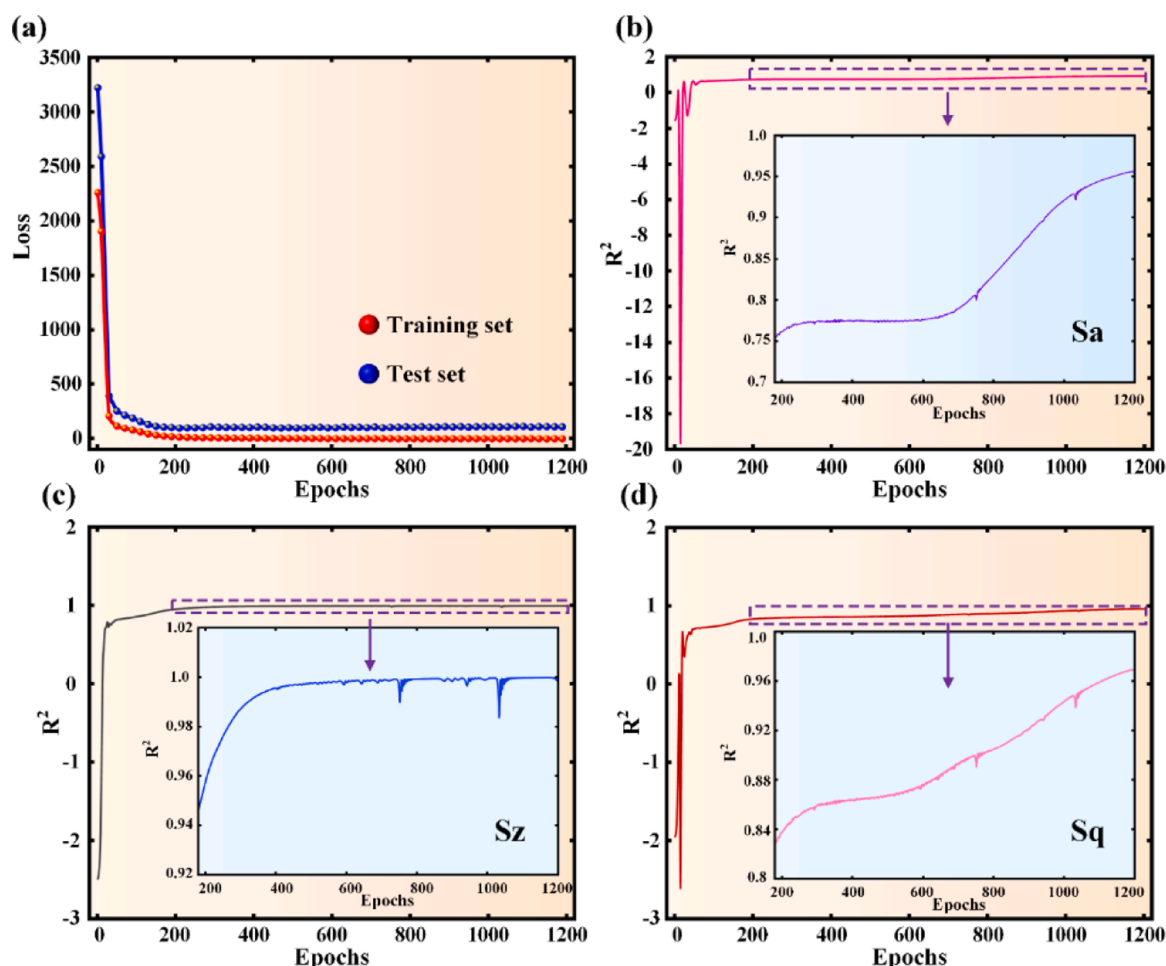


Fig. 6. Variation of the loss function and R^2 value with respect to training round.

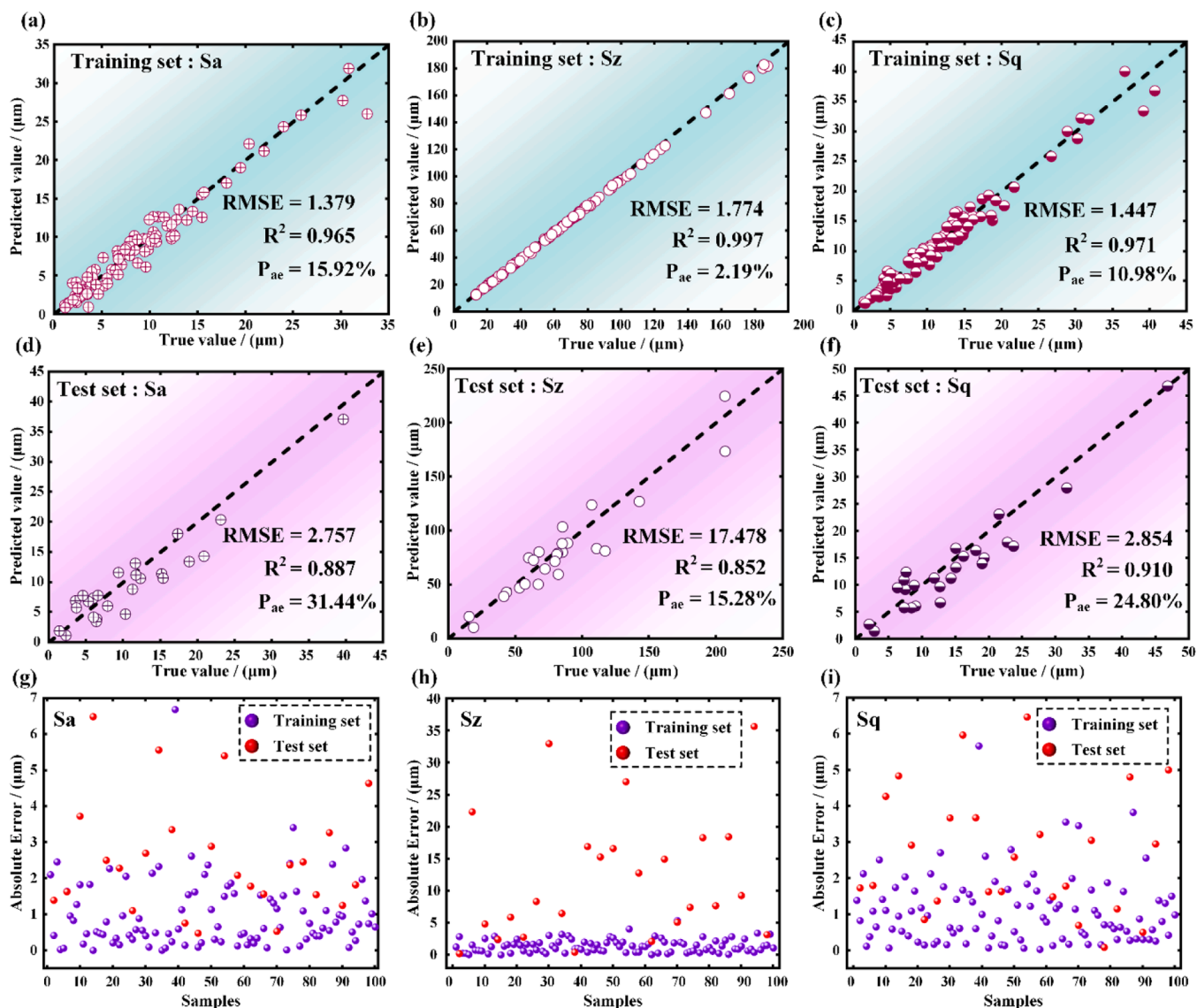


Fig. 7. Prediction results and performance evaluation of the ANN model: The actual and prediction results of the training set: (a) the results of S_a ; (b) the results of S_z ; (c) the results of S_q ; The actual and prediction results of the test set: (d) the results of S_a ; (e) the results of S_z ; (f) the results of S_q ; Results for the performance evaluation indicators: (g) the results of S_a ; (h) the results of S_z ; (i) the results of S_q .

and the RMSE values are 1.379, 1.774 and 1.447 respectively, which appear to be quite small. The prediction results for the test set can be found in Fig. 7d-e. It can be observed that the predicted values of the 25 samples in the test set all exhibit certain deviations compared with the actual values. The R^2 values of all the three roughness parameters are decreased from 0.96 to 0.88. As a matter of fact, this decrease is acceptable, especially for the samples of the test set since it indicates that the no overfitting has occurred for the model after the test process. In addition, the RMSE of S_z has been increased from 1.774 to 17.47, which could be attributed to the large deviation of predicted values for some certain samples due to the relatively small number of samples in the test set. In terms of the P_{ae} results, the S_z possesses a P_{ae} value of 15.28 % which is rather acceptable. As for S_a and S_q , the P_{ae} values of these two parameters are increased to around 25 % for the test set. However, the RMSE values of these two parameters does not show significant increase and still remain at a relatively low level. This should be mainly attributed to the small original values of S_a and S_q , which can drastically magnify the P_{ae} value in the case that deviation exists between the predicted values and actual values. To evaluate the difference between the predicted values and actual values more intuitively, Fig. 7g-

i shows the AE value for every sample in the training set and test set. It can be found that the distribution of the AE values of the training set is quite even. While the AE values for some certain samples are relatively high, and it potentially results in the high P_{ae} values for S_a and S_q . In terms of the test set, the value differences for the predicted values and actual values of S_a and S_q are smaller than 5 for the majority of the samples, demonstrating the superior fitting effect of the model.

During the training process, ANN model utilizes connection layers and activation functions to realize the fitting of complex function relationship among different features. Therefore, the number of connection layer in the ANN model will generate significant impact on the model performance [33]. Fig. 8a-c shows the performance indicator analysis of the ANN model prediction results for the test set with different numbers of connection layer. Controlling the number of connection layer in ANN model plays a key role to guarantee the performance, effectiveness and extendibility of the model. When the number of connection layer is set as 1, the model directly establishes the mapping relationship between the input parameters and output parameters. Under this condition, there is variation of dimensionality, and thus the prediction capability of the model is quite limited. The RMSE value and AE value are 37.56 and

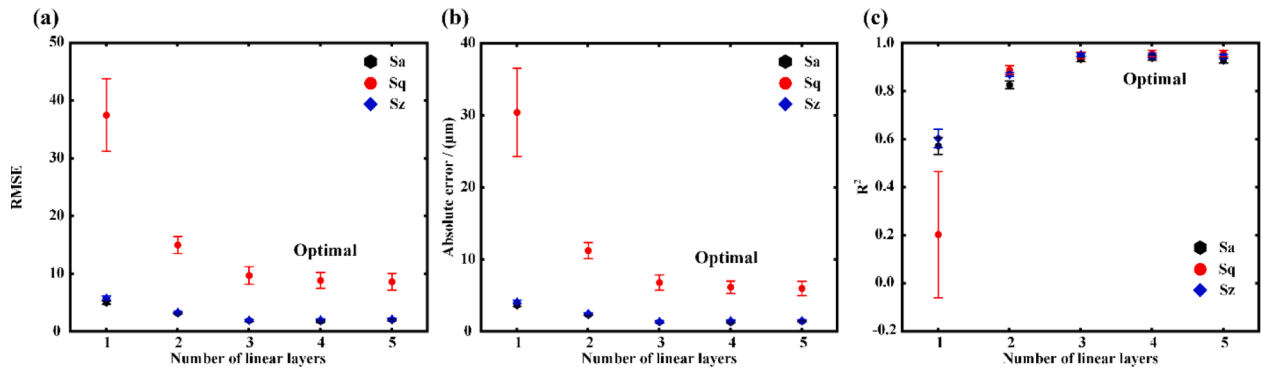


Fig. 8. Evaluation indicator analysis for the model prediction performance of the ANN model with different numbers of layers: (a) RMSE; (b) AE; (c) R^2 .

30.46 μm respectively, and the R^2 value is small as well. Furthermore, with 1 connect layer, the error of evaluation indicators for the model prediction results is large. This is mainly associated with the random partition of the data set, in which a random number is introduced to divide the data set 10 times randomly and the prediction is conducted for the test set. This indicates that with only 1 connection layer, the fitting capability of the model is highly correlated to the partitioning method of the data set, which also demonstrates the mediocre generalization ability of the model. With the increase of the number of connection layer, the RMSE shows a decreasing trend on the whole, indicating the enhancement of model performance. When the number of connection layer is set as 4, the RMSE values of S_a and S_z are decreased to 1.831 and 2.03 respectively, and their AE values are decreased to 1.309 μm and 1.501 μm respectively. The RMSE and AE values of S_z are decreased to 8.866 and 6.186 respectively reaching the lowest point. In addition, as the number of connection layer is 4, the R^2 values of all the roughness parameters are around 0.95, demonstrating that the model can realize accurate prediction of the samples in the test set under this condition. Subsequently, with the increase of the number of connection layer, the error is slightly increased, which can be attributed to over-fitting or increase of computational complexity. Therefore, the number of connection layer for the model is determined as 4.

3.4. Experimental validation of ANN model

The above-mentioned analysis for the model performance was conducted using the laser textured samples treated by the laser processing

parameters within the pre-set process window. To further evaluate the generalization ability of the model, two more samples were prepared using the laser processing parameters beyond the pre-set process window, as shown in Fig. 9. The laser processing parameters used for the sample preparation are: Sample 1 was processed using the pitch of 120 μm , scanning speed of 60 mm/s, laser power of 4.12 W; Sample 2 was processed using the pitch of 180 μm , scanning speed of 180 mm/s, laser power of 6.18 W. It can be found from Fig. 9 that periodic micro-columns with nanoparticles covered on top were well fabricated for both samples, and no distinct cracks were generated among the laser-induced micro-columns. This indicates that the statement claimed in Section 3.1 that both high scanning speed and high laser power are required for the formation of regular periodic micro/nanostructures is quite reasonable. As shown in Figure 9aIII, the predicted surface roughness values S_a , S_q and S_z for Sample 1 are 7.27 μm , 67.14 μm and 10.03 μm respectively, which are quite close to the experimentally measured values (7.96 μm , 65.33 μm and 10.73 μm) and the errors are less than 10 %. For Sample 2 (Figure 9bIII), the prediction percentage errors for S_a and S_q values reach 24.2 % and 12.7 %, while the absolute errors are only 1.02 μm and 0.77 μm . This is mainly attributed to the relatively low values for the original data of S_a and S_q (4.21 μm and 6.06 μm). For S_z , the prediction percentage error is only 6.48 %, and the absolute error is 2.22 μm . This clearly demonstrates that the developed machine learning model could well predict the surface roughness values for the laser textured samples processed using the parameters both with and beyond the process window, and thus exhibits superior generalization ability.

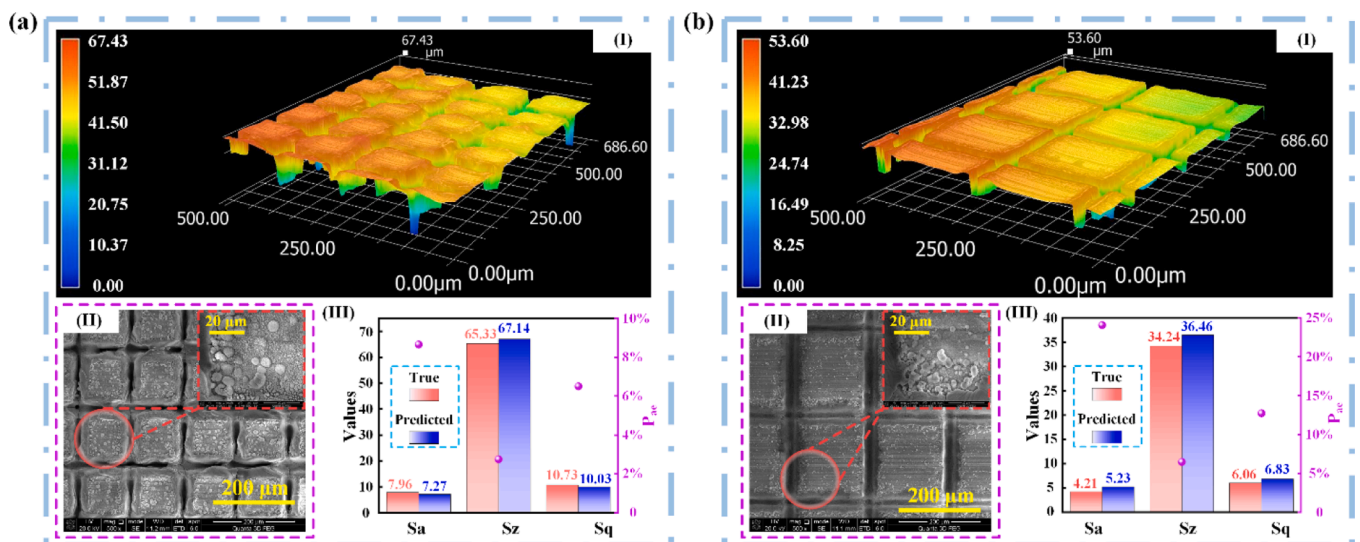


Fig. 9. Surface morphology and prediction result analysis for the laser textured samples processed using the laser processing parameters beyond the pre-set process window: (a) Sample 1; (b) Sample 2.

4. Conclusions

Based on the LASSO regression model and ANN model, this work investigates the effects of laser processing parameters including scanning speed, laser power and pitch on the surface roughness of laser ablated zirconia ceramics surface. The experimental results indicate that different combinations of laser processing parameters will result in diverse types of surface roughness and surface morphology. Therefore, reasonable control of laser processing parameters is critical for ensuring the regularity of the laser-induced periodic surface micro/nano-structures on the zirconia ceramics surface, especially that high power should be accompanied with high scanning speed. For the LASSO model, the prediction results presents good agreements with the actual results for those data at low roughness values on the training set, while the prediction performance deteriorates when the roughness value increases. On the validation set, the prediction accuracy for S_a and S_q is relatively poor, and the R^2 value is only 0.5 and the P_{ae} values are 57.233 % and 41.48 % respectively. However, the prediction result for S_z is relatively good with R^2 value of 0.5 and P_{ae} value of 20.29 %. This reveals its inherent limitations in capturing the complex nonlinear relationship between the laser processing parameters and surface roughness parameters (S_a , S_q , S_z). On the contrary, the ANN model exhibits more improved prediction effect, and the effects of training round and number of linear layers on the model performance is also discussed. The experimental results demonstrates that the ANN model exhibited superior performance, achieving excellent agreement with training data ($R^2 > 0.95$) and maintaining robust predictive accuracy on the validation set with low RMSE and P_{ae} ranging from ~ 15 % (S_z) to ~ 25 % (S_a and S_q). This successful implementation of an ANN represents a significant advancement, effectively addressing the challenge of modeling the intricate non-linear dependencies between the laser parameters and surface roughness. In addition, the ANN model is utilized for prediction of the laser-induced surface roughness prepared using the laser processing parameters beyond the pre-set process window for evaluation of the generalization ability of the model. It is demonstrated that the prediction results of surface roughness are very close to the actual values, and the P_{ae} value for Sample 1 is less than 10 %.

The results indicate that the ANN model proposed in this work can realize accurate prediction for the three types of surface roughness of the laser ablated zirconia ceramics surface. The superior prediction capability can help to achieve accurate control of surface roughness for the laser ablated zirconia ceramics surface, and thus can be applied in diverse areas including biomedicine. Future work should explore the effects of additional laser processing parameters (e.g., pulse duration, wavelength) on the model performance and extend predictions to other surface properties like surface wettability and wear resistance. In addition, it is necessary to apply the model to other materials and conduct comparative studies with other advanced modeling techniques to further enhance the robustness and interpretability of the model.

CRedit authorship contribution statement

Yongzhe Li: Writing – original draft, Validation, Funding acquisition, Formal analysis, Conceptualization. **Jiajun Fu:** Validation, Formal analysis, Data curation. **Runhan Zhao:** Validation, Formal analysis, Data curation. **Chao Liu:** Validation, Formal analysis, Data curation. **Huixin Wang:** Resources, Investigation. **Qinghua Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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